

What Can Time-Series Regressions Tell Us About Policy Counterfactuals?

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Motivation

- ▶ Counterfactual analysis is at the heart of macroeconomic research.
- ▶ Since Lucas (1976), conventional wisdom among macroeconomists says that valid policy counterfactuals require a **fully-specified structural** model.
 - Empirical relationship is **not** policy invariant.
- ▶ Can we still derive meaningful conclusions about policy counterfactuals from existing **reduced-form** evidences?

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 1. The model is **linear**.
 2. Private sector behavior depends on policy **only through the expected path of the policy instrument**.
 3. We know the dynamic causal effects of **all possible contemporaneous and news shocks** to the prevailing policy rule.
- ▶ **Idea:** replicate alternative policy rule using policy shocks

This paper

- ▶ Characterizes the information needed to compute valid policy counterfactuals.
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 1. Best approximation
 2. Additional assumptions on the model

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- ▶ Presents two practical approaches to implement the theoretical insights under realistic information constraints:
 1. Best approximation
 2. Additional assumptions on the model
- ▶ Shows how to use the characterization to compute optimal policy.

Key idea

- ▶ 2x2 example:

$$\underbrace{\begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}}_{model} \begin{bmatrix} y \\ i \end{bmatrix} = \underbrace{\begin{bmatrix} \varepsilon \\ \nu \end{bmatrix}}_{shock}$$

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- Dynare:

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- The dynamic causal effect Θ , in principle, can be estimated with reduced-form methods.

Key idea

- ▶ Counterfactual policy rule:

$$\begin{bmatrix} 1 & 2 \\ \textcolor{red}{2} & 1 \end{bmatrix} \begin{bmatrix} y \\ i \end{bmatrix} = \begin{bmatrix} \varepsilon \\ \textcolor{red}{0} \end{bmatrix}$$

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- ▶ The linear system becomes

$$\begin{bmatrix} 1 & 2 \\ \textcolor{red}{1} & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} y \\ i \end{bmatrix} = \begin{bmatrix} \varepsilon \\ \textcolor{red}{\nu} \\ \textcolor{red}{\nu} \end{bmatrix}$$

Key idea

► Note that

$$\begin{bmatrix} 1 & 2 \\ 1 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} y \\ i \end{bmatrix} = \begin{bmatrix} \varepsilon \\ \nu \\ \nu \end{bmatrix} \Leftrightarrow \begin{bmatrix} 1 & 2 \\ 1 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} y \\ i \end{bmatrix} = \begin{bmatrix} \varepsilon \\ \nu \\ 0 \end{bmatrix}$$
$$\Leftrightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} y \\ i \end{bmatrix} = \begin{bmatrix} \Theta_s \\ 0 \end{bmatrix}$$

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- Knowledge of Θ is sufficient for computing the policy counterfactuals under the new rule.
- **Intuition:** alternative policy rule is equivalent to prevailing rule + engineered shocks

Notations

- Consider a linearized perfect foresight model economy:

$$\underbrace{\begin{pmatrix} \mathcal{H}_w & \mathcal{H}_x & \mathcal{H}_z \\ 0 & \mathcal{A}_x & \mathcal{A}_z \end{pmatrix}}_{\equiv \mathcal{B}} \begin{pmatrix} w \\ x \\ z \end{pmatrix} = - \begin{pmatrix} \mathcal{H}_\varepsilon & 0 \\ 0 & I \end{pmatrix} \begin{pmatrix} \varepsilon \\ \nu \end{pmatrix}$$

- Define

$$\Theta_{\mathcal{A}} \equiv \begin{pmatrix} \Theta_{w,\varepsilon,\mathcal{A}} & \Theta_{w,\nu,\mathcal{A}} \\ \Theta_{x,\varepsilon,\mathcal{A}} & \Theta_{x,\nu,\mathcal{A}} \\ \Theta_{z,\varepsilon,\mathcal{A}} & \Theta_{z,\nu,\mathcal{A}} \end{pmatrix} \equiv -\mathcal{B}^{-1} \begin{pmatrix} \mathcal{H}_\varepsilon & 0 \\ 0 & I \end{pmatrix}$$

- Define $\{\tilde{\mathcal{B}}, \Theta_{\tilde{\mathcal{A}}}\}$ as the corresponding matrices under an alternative policy rule $\{\tilde{\mathcal{A}}_x, \tilde{\mathcal{A}}_z\}$.

Main result - identification theorem

Proposition. Suppose that $\{\Theta_{x,\nu,\mathcal{A}}, \Theta_{z,\nu,\mathcal{A}}\}$ and $\{x_{\mathcal{A}}(\varepsilon), z_{\mathcal{A}}(\varepsilon)\}$ are known. Then, for any alternative policy rule $\{\tilde{\mathcal{A}}_x, \tilde{\mathcal{A}}_z\}$ that induces a unique, determinate equilibrium, we can recover the policy counterfactuals $\{x_{\tilde{\mathcal{A}}}(\varepsilon), z_{\tilde{\mathcal{A}}}(\varepsilon)\}$ as the unique solution to the system

$$\begin{pmatrix} I & 0 & -\Theta_{x,\nu,\mathcal{A}} \\ 0 & I & -\Theta_{z,\nu,\mathcal{A}} \\ \tilde{\mathcal{A}}_x & \tilde{\mathcal{A}}_z & 0 \end{pmatrix} \begin{pmatrix} x \\ z \\ \nu \end{pmatrix} = \begin{pmatrix} x_{\mathcal{A}}(\varepsilon) \\ z_{\mathcal{A}}(\varepsilon) \\ 0 \end{pmatrix}$$

- $\{x_{\mathcal{A}}(\varepsilon), z_{\mathcal{A}}(\varepsilon)\} = \{\Theta_{x,\varepsilon,\mathcal{A}} \times \varepsilon, \Theta_{z,\varepsilon,\mathcal{A}} \times \varepsilon\}$ is the dynamic causal effects of the shock sequence ε under the baseline policy rule.

Quantitative illustration

ALTERNATIVE POLICY RULE, HANK MODEL

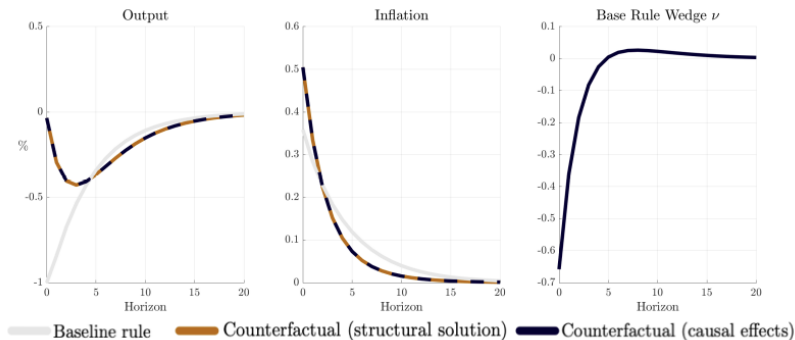


Figure 1: Output and inflation impulse responses together with the equivalence shock wedge ν (see (22)) for the HANK model with policy rules (37) and (38). The impact output contraction under the prevailing baseline rule is normalized to -1% .

- The information requirement is unrealistic: need to know $\{\Theta_{x,\nu,\mathcal{A}}, \Theta_{z,\nu,\mathcal{A}}\}$ i.e. the dynamic causal effects of all possible contemporaneous and news shocks to the prevailing policy rule.

Caveats

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- ▶ A policy rule involves infinitely many (one per period) equations, so we need infinite degrees of freedom to replicate.
 - But in practice, the alternative policy rule may lie in some finite-dimensional subspace.
- ▶ If the alternative policy rule changes the steady-state of the economy, the method is invalid as the $\mathcal{H}$ matrices also change.

Counterfactual rules in identified subspaces

- ▶ Let $\mathcal{V}$ be a matrix whose columns collect the responses of the policy shock ν to $n_{\tilde{\nu}}$ identified sources of variation in policy. (e.g. Gertler & Karadi 2015, Romer & Romer 2004)
- ▶ Implementable shocks are $\nu = \mathcal{V} \times \tilde{\nu}$, where $\tilde{\nu} \in \mathbb{R}^{n_{\tilde{\nu}}}$. Denote the dynamic causal effects of these shocks by $\{\Theta_{x,\tilde{\nu},\mathcal{A}}, \Theta_{z,\tilde{\nu},\mathcal{A}}\}$

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- ▶ An alternative policy rule is replicable if for some $\tilde{\nu} \in \mathbb{R}^{n_{\tilde{\nu}}}$,

$$\tilde{\mathcal{A}}_x(x_{\mathcal{A}}(\varepsilon) + \Theta_{x,\tilde{\nu},\mathcal{A}} \times \tilde{\nu}) + \tilde{\mathcal{A}}_z(z_{\mathcal{A}}(\varepsilon) + \Theta_{z,\tilde{\nu},\mathcal{A}} \times \tilde{\nu}) = 0$$

If it is not replicable, we can always find the least norm solution (i.e. linear regression).

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- ▶ Use the investment-specific technology shock from Ben Zeev & Khan (2015) and the MP shocks from Gertler & Karadi (2015) and Christiano, Eichenbaum, Evans (1999).
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- ▶ Compute the best-approximated policy counterfactuals using the estimated dynamic causal effects.

POLICY COUNTERFACTUAL FOR INVESTMENT SHOCKS, IDENTIFIED SUBSPACE

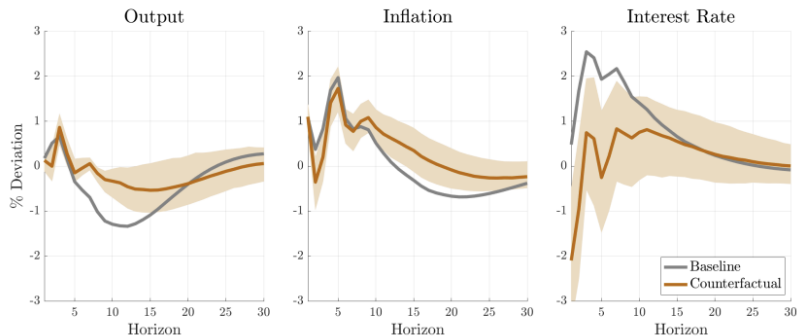


Figure 3: Output, inflation and interest rate impulse responses to a contractionary investment-specific technology shock under the prevailing baseline rule (grey) and the best feasible approximation to a rule that stabilizes output (orange). The shaded areas correspond to 16th and 84th percentile confidence bands.

Additional identifying information

- ▶ Another approach is to impose (dogmatic) prior on the impulse response matrix $\Theta_{q,\nu,\mathcal{A}}$.
- ▶ To compute policy counterfactuals, Sims & Zha (2006) proposes to subject the economy to a sequence of *unanticipated* policy shocks that enforces the alternative policy rule along the equilibrium path.
- ▶ By the identification theorem, their approach is valid if

$$\Theta_{q,\nu,\mathcal{A}} = \begin{pmatrix} \Theta_{q,\nu,\mathcal{A}}(1,1) & 0 & 0 & \dots \\ \Theta_{q,\nu,\mathcal{A}}(2,1) & \Theta_{q,\nu,\mathcal{A}}(1,1) & 0 & \dots \\ \Theta_{q,\nu,\mathcal{A}}(3,1) & \Theta_{q,\nu,\mathcal{A}}(2,1) & \Theta_{q,\nu,\mathcal{A}}(1,1) & \dots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

That is, news shocks have no effect.

Prior on the model

- ▶ How can prior on some blocks of the model help identification?
- ▶ NKPC:

$$\boldsymbol{\pi} = \Pi_y \times \mathbf{y} + \Pi_\varepsilon \times \boldsymbol{\varepsilon}$$
$$\Pi_y = \begin{pmatrix} \kappa & \kappa\beta & \kappa\beta^2 & \dots \\ 0 & \kappa & \kappa\beta & \dots \\ 0 & 0 & \kappa & \dots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

- ▶ NKPC implies

$$\Theta_{\pi, \nu, \mathcal{A}} = \Pi_y \times \Theta_{y, \nu, \mathcal{A}}$$

Incorporating the output-inflation prior

- Observe that the identification system

$$\begin{pmatrix} I & 0 & -\Theta_{y,\nu,\mathcal{A}} \\ 0 & I & -\Theta_{\pi,\nu,\mathcal{A}} \\ \tilde{\mathcal{A}}_y & \tilde{\mathcal{A}}_\pi & 0 \end{pmatrix} \begin{pmatrix} y \\ \pi \\ \nu \end{pmatrix} = \begin{pmatrix} y_{\mathcal{A}}(\varepsilon) \\ \pi_{\mathcal{A}}(\varepsilon) \\ 0 \end{pmatrix}$$

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is equivalent to

$$\begin{pmatrix} I & 0 & -I \\ 0 & I & -\Pi_y \\ \tilde{\mathcal{A}}_y & \tilde{\mathcal{A}}_\pi & 0 \end{pmatrix} \begin{pmatrix} y \\ \pi \\ \nu' \end{pmatrix} = \begin{pmatrix} y_{\mathcal{A}}(\varepsilon) \\ \pi_{\mathcal{A}}(\varepsilon) \\ 0 \end{pmatrix}$$

by change of variable $\nu' \equiv \Theta_{y,\nu,\mathcal{A}} \times \nu$.

- Need to assume that the map $\Theta_{y,\nu,\mathcal{A}}$ is invertible, which is true in standard business-cycle models.

- ▶ Assume a hybrid Phillips curve relationship:

$$\pi_t = \gamma_b \pi_{t-1}^4 + \gamma_f \mathbb{E}_t[\pi_{t+4}^4] + \kappa y_t + \varepsilon_t$$

where $\pi_{t-1}^4 = \frac{1}{4} \times (\pi_{t-1} + \pi_{t-2} + \pi_{t-3} + \pi_{t-4})$.

- ▶ Estimate the parameters $\{\gamma_b, \gamma_f, \kappa\}$ (and so the matrix Π_y) using estimated dynamic causal effects of identified monetary shocks. (Barnichon & Mesters 2020)
- ▶ Compute the policy counterfactuals using Π_y .

POLICY COUNTERFACTUALS FOR INVESTMENT SHOCKS, PC EXTRAPOLATION

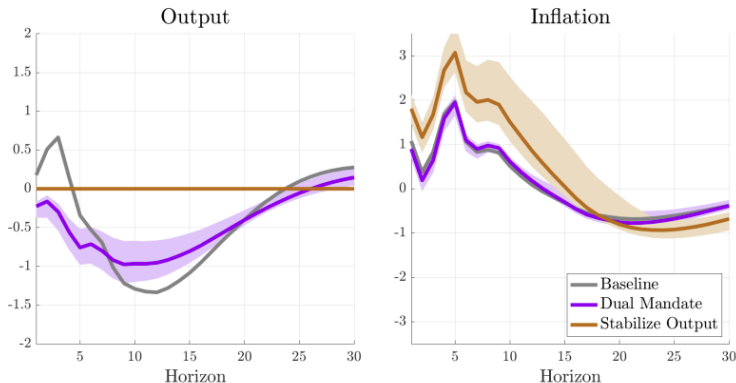


Figure 5: Output and inflation impulse responses to a contractionary investment-specific technology shock under the prevailing baseline rule (grey), a counterfactual rule that perfectly stabilizes output (grey), and the optimal policy rule for a dual-mandate policymaker ($\lambda_{\pi} = \lambda_y = 1$) (purple). The shaded areas correspond to 16th and 84th percentile confidence bands.

Optimal policy

- ▶ The method can also be used to compute optimal policy for a quadratic loss function of the form

$$\mathcal{L} = \sum_{i=1}^{n_x} \lambda_i x_i' W x_i$$

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- ▶ **Solution:**

$$\begin{aligned}\mathcal{A}_x^* &= (\lambda_1 \Theta'_{x_1, \nu, \mathcal{A}} W, \dots, \lambda_{n_x} \Theta'_{x_{n_x}, \nu, \mathcal{A}} W) \\ \mathcal{A}_z^* &= 0\end{aligned}$$

- ▶ With a sufficiently large menu of identified policy shocks, we can compute valid policy counterfactuals without fully specifying a structural model.
- ▶ The method can also be used as a numerical trick. See Groot et al. (2021) and Hebden & Winker (2021).

Some thoughts

- ▶ Hard to interpret the policy counterfactuals – the domain on which the counterfactuals are valid is very unclear.
 - Very often shocks are identified by structural assumptions. The domain shrinks whenever we add these shocks to our menu.
 - Even worse, what if those structural assumptions are not consistent with each other?

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 - Very often shocks are identified by structural assumptions. The domain shrinks whenever we add these shocks to our menu.
 - Even worse, what if those structural assumptions are not consistent with each other?
- ▶ The best approximation approach has no theoretical ground.
 - Why should one prefer this approximated counterfactual to a counterfactual computed from a fully specified model?

► Model equations:

$$y = C_y y + C_\pi \pi + C_i i + C_\tau \tau + \varepsilon^d$$

$$\pi = \kappa y + \beta \pi_{+1} + \varepsilon^s$$

$$0 = \bar{\tau} \tau + \bar{b}(1 + \bar{i})(i_{-1} - \pi)$$

$$i = \phi_i i_{-1} + (1 - \phi_i)(\phi_\pi \pi + \phi_y y) + \nu$$

► Baseline: $\phi_i = 0$, $\phi_\pi = 1.5$, $\phi_y = 0$

$$i_t = \phi_\pi \pi_t$$

► Alternative: $\phi_i = 0.9$, $\phi_\pi = 2$, $\phi_y = 0.5$

$$i_t = \phi_i i_{t-1} + (1 - \phi_i)(\phi_\pi \pi_t + \phi_y y_t)$$

Monetary policy shocks

FEDERAL FUNDS RATE, CHRISTIANO ET AL. (1999) VS. GERTLER & KARADI (2015)

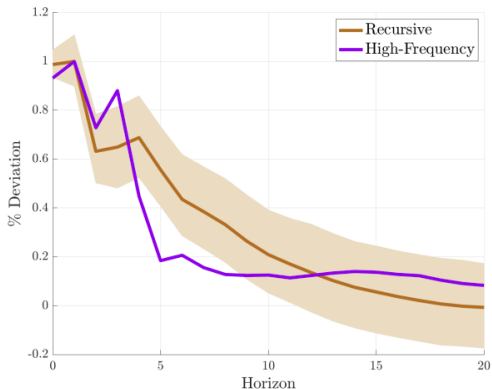


Figure B.1: Federal funds rate impulses to the recursive (orange) and high-frequency (purple) monetary policy shocks of Christiano et al. (1999) and Gertler & Karadi (2015), with the peak impulse normalized to 1 per cent.